How Spatial inequalities contributed to the uneven spread of COVID-19 in South Africa, Kenya and Ghana

The effects of the Pandemic on inequality and poverty

Prior to the pandemic, Lakner et al (2022) research indicated that more than half of the population living in extreme poverty are situated in Sub-Saharan Africa. Poverty is expected to see increases at a faster rate than before the Pandemic. These countries are significantly more vulnerable to health and economic crises such as COVID-19, largely due to less-established and weak economic structures and health systems in place. By contrast, developed nations have much more security in terms of well-developed financial and economic systems and resources to shield against any proverbial health/economic storm. Therefore, the question still remains of whether adopting high-income country's prevention policies for developing nations ensures the well-being of those who live in regions with high levels of spatial inequalities.

COVID prevention policies implemented in countries with high income inequality, high unemployment, and low economic growth, will produce skewed outcomes compared to developed countries. Most developing countries' citizens live in regions with little access to water, safe sanitation and reliable electricity. Therefore, in the attempts to protect citizens from becoming infected, enforced lockdowns in low-income nations did not produce the same results as high income countries, since being confined to households without clean water, safe sanitation and reliable electricity increased exposure and risk of infection.

Struggling economic hubs were brought to a halt, having further profound effects on rural and low-income regions with pre-existing social and spatial inequalities. These inequalities were exposed and reinforced, and is one of the primary causes of the uneven spread of the virus across developed and developing nations, since poorer regions were not lockdown ready and thus more vulnerable.

Poverty, hunger and food insecurity rates have increased in developing African nations, and will continue rising. Historically, past epidemics have contributed to consistent increases poverty and social inequality levels, even 5 years after the epidemics.

Additionally, increases in unemployment rates and decreases in income has significantly contributed to widening the expanses of social and income inequalities present in such countries.

The Pandemic and Attributes of Developing Nations

High levels of social and spatial imbalances are commonplace in developing and low-income nations. Clean piped water, safe sanitation and electricity is not always accessible to those in developing nations. Lockdowns significantly limit citizens' mobility and daily economic activities and would

substantially result in reduced access to food, reduced monthly income and reduced labour forces. Such regions, are mostly not in the position to endure financial and economic shocks, usually resulting in lingering and multiplied hardships for the poor and most vulnerable.

Developing nations are characterized by high levels of pre-existing social and spatial inequalities, particularly in rural regions. Employing prevention policies that are applicable to high-income countries, in these nations, will produce different outcomes. COVID vulnerability and virus lethality will differ between those who have access to wealth and assets and those who do not. Vulnerability and lethality will also differ between those with better living conditions and those living in poorer conditions. Moreover, underlying health risk factors are more likely to be present in poorer communities than the rich, influencing overall vulnerability.

World Bank (2018) Inequality research pertaining to the Southern African Customs Union shows that increased inequalities in countries such as South Africa, are attributed to the poor functioning of urban labour markets, the present gender earnings gap and constrained rural land markets. Furthermore, place-based disadvantages and the inability of a child to break out from poverty further entrenches inequality, in that the chances of the child achieving a higher standard of living than the household in which they were reared is very low.

Current demographic, environmental, socio-economic and political conditions were shown to have had a significant impact on countries' COVID burdens (Manda, Darikwa, Nkwenika and Bergquist, 2021). Low-income countries with poor conditions struggled to stay afloat during the throes of the Pandemic, compared to richer countries. Countries with high levels of pre-existing inequalities and income inequality were at a significant disadvantage prior to the Pandemic, as those with high levels of socio-economic inequalities were more likely to be less lockdown prepared and therefore more vulnerable.

The relationship between lockdown preparedness, COVID-19 vulnerability and wealth status: South Africa, Kenya and Ghana

Through examining the relationship between lockdown preparedness, COVID vulnerability and wealth status, regions that are relatively less lockdown ready and more vulnerable to infection are identified. This aids in directing a greater level of support and intervention to ensure the survival of the most vulnerable. There is a need for greater government support to be extended to those in poor rural regions and informal settlements located in urban regions. Government must improve the poor's access to basic necessities and services, enabling the poor to become more lockdown ready and less vulnerable, should future outbreaks occur.

Global measures intended to limit the spread, such as social distancing and hand hygiene, did not succeed in poorer regions due to these areas being densely populated and associated with little access to clean water, hand sanitizer and soap.

Owoo et al (2021) defines lockdown preparedness as households' living conditions that would allow for strict lockdown within a country. Figure 1 highlights these conditions. Meeting all conditions stated

in Figure 1 is considered fully ready, while meeting the first three conditions is considered partial lockdown readiness.

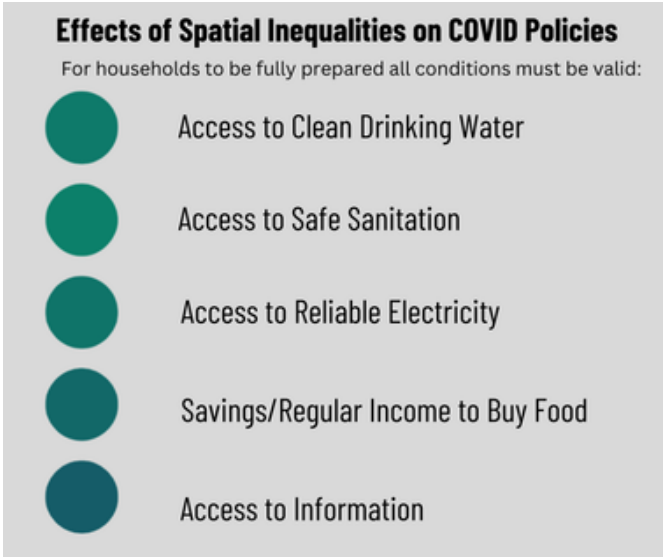


Figure 1

Vulnerability to infection is defined as a household's risk of becoming infected due to living conditions and interactions with individuals outside of the household. Vulnerability indicators would relate the ability of individuals to follow WHO guidelines. Common vulnerability indicators are shown in Table 1.

COVID vulnerability intensity is higher where there is a high number of valid vulnerability indicators and lower wealth scores present. Further, intensity is likely to be higher in households with more elderly people, since they are more likely to have underlying health conditions, such as chronic diseases, food insecurity and lower

Vulnerability Indicators

Indicator	Scientific Reasoning
People older than 60 years living in Households	People that are 60 years are older are at higher risk of becoming severely infected and dying.
Large Households - more than 6 people	An infected person is more likely to infect their household members
Sharing Water and Toilet Facilities	The Sharing of water and toilet facilities increase the risks of contracting COVID between neighboring households
No refrigerator	Lack of refrigeration increases the number of trips to obtain food
No access to radio/tv	Lack of reliable information channels could lead to increased vulnerability

Adapted from Manda et al, (2021)

income.

Table 1

Wealth Status: A key determinant of Lockdown preparedness and COVID vulnerability

A household is able to protect themselves from infection and endure lockdowns as far as wealth and income status permits. Wealth status is a key determinant of households' ability to be lockdown prepared and to protect itself from the virus. This is empirically confirmed by the significant, positive relationship between wealth status and lockdown preparedness and the negative, significant relationship between COVID vulnerability and wealth scores. Those with little access to material wealth and assets would fare poorly in lockdown preparedness and vulnerability.

Wealth status is relatively rigid in the short to medium term and is unlikely to change very rapidly in the wake of crises such as COVID. As in most developing nations, the virus affected those in poorer regions disproportionately compared to wealthier regions. The most vulnerable did not have adequate means to cope with extended lockdowns, as the lack of wealth results in poor living conditions where basic amenities are oftentimes denied. The poor have little to no access to clean piped water in their homes, leading them to collect water from water sharing points. This increases the rate of exposure between those that collect water from communal points. The absence of basic needs and services in poor regions, is a major underlying factor contributing to spatial differences in vulnerabilities. Additionally, poorer citizens can generally be associated with weaker immune systems and overall health, rendering them more vulnerable to exposure and infection.

South Africa

South Africa's history has contributed to making it one of the most income unequal countries in the world, with an income Gini-coefficient of 0.67. Socio-economic and spatial inequalities continue to persist despite significant progress in service delivery. However, the appearance of the Pandemic exposed and reinforced these inequalities all the more.

Shifa, David and Liebbrandt (2020), find that initial levels of inequality determines the level of virus exposure and its lethality, leading to significant inequalities in COVID vulnerability. Moreover, findings show stark spatial inequalities in vulnerabilities in South Africa, with actual vulnerabilities being more pronounced for the poor, since those that lack essential amenities were not lockdown prepared and therefore more vulnerable. Poor South African citizens were unable to comply with preventative measures irrespective of where they lived, rendering them unable to protect themselves from the virus partly due to their wealth status, lack of basic amenities and living conditions.

Key demographics highlighted in the study, show that at least 21% of the population live in a household with at least 1 elderly person and two younger individuals. 15% of the population did not have access to a refrigerator, while almost 8% could not access a television or radio. Presence or absence of these factors put this proportion of the population at higher risk of infection.

Vulnerability and Vulnerability Intensity

Eastern Cape, Limpopo and KwaZulu-Natal had the highest average vulnerability scores. In the Eastern Cape and Limpopo, the proportion of the population sharing water is more than 50%, with almost 30% of the Eastern Cape population lacking access to a refrigerator. For other regions such as Gauteng and Western Cape, water sharing is less than 10% and 8%, respectively. Eastern Cape and KZN provinces contain 34 of the poorest municipalities. Vulnerability is disproportionately higher in the poorest wealth quintiles in all provinces, due to poor living conditions, as these citizens' home environments do not enable them to comply with adequate hand hygiene and social distancing. Additionally, the intensity of vulnerability is higher in the aforementioned provinces with the highest average vulnerability scores, including the North West. Despite Limpopo having a lower population share, it still accounts for a high vulnerability national share. Two key observations show that even within regions where vulnerability is low, there are locations that are highly vulnerable. Despite high vulnerability in certain provinces, high vulnerability paired with large population is relatively limited.

Statistics SA examines households' affordability thresholds, finding that 59% of all households fall below the income threshold of R2300 per month. The equitable share formula that aids municipalities in providing free basic water, sanitation, electricity and refuse collection services was established to serve those living below this threshold. By availing these services to the poor, overall vulnerability will significantly improve, as these basic services are closely aligned with vulnerability indicators.

An Internet article published by Relief Web, reported on the approval of the $750 million World Bank Development Policy Loan extended to South Africa. Funds were directed toward helping the poor and the vulnerable against the adverse effects of the Pandemic. The R350 Social Relief of Distress (SRD) Grant that was disbursed to the poor and vulnerable during the Pandemic was extended to March 2024.

Kenya

Using data from the Kenyan Household budget survey and the National census, Damiano-Kolundo, et al (2021) found that during the 2 documented years of study, both lockdown readiness and vulnerability increased. Readiness scores were higher in urban regions than rural regions, while vulnerability was higher in rural regions compared to urban areas. At the national level, lockdown readiness did improve, with pronounced improvement occurring in rural regions.

Northern, Eastern and Western regions of Kenya were found to be less lockdown ready compared to the Coastal rift valley, Central and Southern Kenyan regions. Save for major urban regions and central parts of Kenya, statistics show that counties have higher proportions of households with larger families with presence of elderly people, thereby increasing vulnerability within households. Over 2 years, improvement was seen in the populations' access to tv/radio, electricity and sanitation facilities that were not shared. However, there was a decline in access to safe water and formal employment. Rural regions had more water collection points compared to urban counties, thereby increasing the

risk of exposure and infection. Findings show that more than 50% of households shared water sources and toilets in both years under study.

The proportion of households that were fully lockdown ready was higher than partially ready households, common in urban regions. Both partially and fully ready households were higher in urban regions, in the cases of both male and female headed households.

Vulnerability and Vulnerability Intensity

Over the 2 year study, COVID vulnerability increased for both rural and urban regions. Inequality levels were higher for urban regions compared to rural regions prior to 2016, with these differences between regions narrowing by 2019. Furthermore, female-headed households were more vulnerable compared to male-headed households at the start of the study period, eventually with marginal difference by 2019. Between 8%-9% of the population were vulnerable due to the presence of 1 vulnerability indicator, while for two and three indicators, this figure stood at roughly 30% and 40%, respectively.

20% of the population were vulnerable corresponding to those who presented with 4 vulnerability indicators.

Up to the first 2 vulnerability indicators, intensity of vulnerability is higher in urban regions, with rural regions' vulnerability intensifying thereafter. Households vulnerable to more indicators did however decrease over this 2 year period.

In terms of the influences that wealth effects have on lockdown preparedness and vulnerability, the Kenya study echoes South African findings, indicating that lockdown preparedness increases with wealth scores, while COVID vulnerability decreases with increases in wealth.

To shield the most vulnerable, the Kenyan Government made USD $95 million available in cash transfers for poor citizens and the elderly. Tax relief measures were implemented for those earning below a certain income bracket, while the Central Bank lowered the Central Bank Rate by 1 percent, as well as other multiple government interventions.

Ghana

Analyzing data from the Ghana Socio-economic panel survey, Owoo et al (2021) finds that at a national level only 10% are lockdown ready, with more fully lockdown ready households situated in urban regions. 0.5% of the rural households were found to be ready for lockdown compared to 7.4% in urban regions.

Significant spatial and locational differences concerning lockdown preparedness and vulnerability was evident. Greater vulnerabilities in secondary infection rates were common in poorer, rural areas, especially in Northern parts of Ghana. The study outlines the negative relationship between a households lockdown preparedness and its associated COVID vulnerability, being that the higher the households lockdown preparedness, the lower is that households vulnerability.

Poorer, male-headed households were found to be poorly lockdown prepared, especially in rural and Northern regions of Ghana. Southern Ghana was more lockdown prepared compared to Northern regions.

As in most cases, spatial differences in lockdown preparedness and COVID vulnerabilities is primarily due to spatial inequalities, such as inaccessibility to clean water, safe sanitation and reliable electricity.

41% of households had access to clean water according to the broader water definition, including water from water tankers and sachet water, with 16% of the population having access to safe sanitation. 2% of Ghana's poor owned assets such as fridges compared to 77% of the richest population, indicating the positive relationship between asset wealth status and lockdown preparedness. Over a third of households from the richest populations are fully lockdown ready, with no households in the poorest wealth quintile being fully prepared. For the second and third wealth quintile, more than 1% and 4% were fully prepared, respectively.

Vulnerability and Vulnerability Intensity

Covid vulnerability was found to be higher and more concentrated in poorer, rural and male-headed households in Ghana. Households in rural regions proved more vulnerable to exposure and infection since they were not adequately lockdown prepared. 35% of households located in rural Ghana are vulnerable to 4 indicators, compared to 11% in urban households. 0.5% of households in rural Ghana were found to be vulnerable in all stated indicators, compared to just 0.1% in urban Ghana.

A greater proportion of female-headed households were not vulnerable to all vulnerability indicators, compared to male-headed households. This is attributed to female-headed households being more likely to have access to clean water, safe sanitation and reliable electricity.

Households with more than 6 members were mostly located in the Northern regions where vulnerability scores were high and preparedness was low. Vulnerability was found to be higher in households with elderly members, common in the Upper East and Greater Ghana Regions.

Due to the larger population, the Ashanti region contributes a quarter to the national share that is vulnerable to one indicator. Similarly, the Northern region accounts for 26% of the national share that is vulnerable in 5 indicators. Highest average vulnerability intensity scores occur in Northern Ghana (37%) and the upper west (19%). The Northern, Upper East and Upper West regions experienced the highest level of vulnerability intensity, including the Brong Ahafo region.

Influences of access to assets on Lockdown readiness and COVID Vulnerability

The effects of the Pandemic resulted in serious economic consequences. Labour income shocks, a reduction in the labour force and number of hours worked, collectively imposed significant income effects on the poor and vulnerable. Global consequences included the decline of global GDP and

growth forecasts which were directly passed through to individual households incomes, where falling income increased overall vulnerability and reduced lockdown readiness levels. Those with little or no income are usually those that do not remain standing in times of economic distress. Poor and rural regions are more likely to have higher unemployment rates amongst other adversities which disable citizens shielding themselves from such economic shocks. For this reason, extreme poverty cases increased drastically across developing countries.

The population with access to assets and wealth are more likely to have acceptable living conditions, enabling them to be more lockdown ready, lowering vulnerability.

Findings in Ghana, Kenya and South Africa all attest to the reality that those with little or no access to assets and general material resources were associated with higher vulnerability rates and lower lockdown preparedness. For South Africa, 30% of the population with no access to assets were relatively more vulnerable. The figure was less than 2% for the richest population. Similar results occur in Ghana. In Kenya, only 16% in the poorest wealth bracket were lockdown ready, compared to 68% in the richest bracket. Vulnerability scores for the poorest and richest wealth brackets were 59% and 34%, respectively.

To highlight these pronounced effects on the disadvantaged, low-income citizens with relatively high income uncertainty were four times more likely to lose their livelihoods as a result of the pandemic, consequently causing the reverse-migration of urban dwellers back to rural areas for survival. Therefore, the access to assets, wealth and secured income are key determinants of a households lockdown readiness and vulnerability.

Necessary Government Solutions

Fiscal policy can reduce income inequality and poverty. Through prioritizing and addressing its historical and economic origins, government can enable the most vulnerable to endure crises such as the Pandemic. Government interventions need to reach even those in informal settlements located within low-vulnerability urban areas, which requires both national and local institutional capacities. Poorer regions would benefit immensely through improved accessibility to clean water, safe sanitation and reliable access to electricity and refrigeration facilities, resulting in reduced COVID vulnerability and increased Lockdown preparedness.

Shifa, et al (2021) states the importance of explicitly identifying spatial concepts embedded within the design of Pandemic policy responses in future. Spatial targeting policies can be implemented in the plight against income inequality and poverty. Through the improving of provision to basic amenities, improving hunger and food insecurity programs and social protection programs and sustained investment in education, healthcare and job creation, government can prioritize support for those municipalities and regions that require urgent aid.

Provision of Basic necessities and services

Interventions in terms of policy solutions are required to shield the poor and vulnerable from future health and economic crises. Basic services accessible to every citizen is most important to ensure household readiness and lower vulnerability. For example, clean piped water with limited sharing points and safe sanitation facilities can reduce risk of exposure and infection significantly. Making regular income available for food and essential purchases is another measure to increase these citizens' protection. Further, access to reliable information channels can also aid in shielding and preserving the lives of the nations most vulnerable.

Hunger and Food Insecurity

Government regulations can play a significant role in ensuring secure access to food. Perhaps not a direct influence in this plight, however, through legislation and partnership development initiatives, government can assist in ensuring the basic rights to access to food and water to those living in poverty. In order to reduce pressures on government to assist in this way perpetually, reducing unemployment and regulating staple food price increases can position the most vulnerable to withstand economic and health turmoil. This can be accomplished through food subsidies and the strengthening of relationships with external partners and charitable initiatives.

Social Protection Policies

Approximately the income and consumption of 3.2 million people will have contracted by 10%, directly increasing poverty. World Bank research asserts that social protection programs have the potential to cushion 2/3 of the most vulnerable from this impact. Increased allocations to existing social welfare programs need attention. Further, adaptive social welfare programs can be instrumental in shielding the most vulnerable from economic and financial shocks. Further, reliability of State support is required, with efforts to reduce current government shortcomings and oversights.

Continuous Investment in Healthcare, Education, and Job Creation

Over the next 5 years, effects of the Pandemic are expected to increase mortality rates for those with HIV, TB and Malaria patients in Africa. Continuous investment and improvement of current healthcare systems can shoulder future health burdens in developing nations. The same can be said of education accessibility and quality in developing nations. Education would increase the chances of those most vulnerable to acquire jobs that can be performed remotely, aiding in the lockdown preparedness.

Small and medium enterprises are the driving forces of economic growth, and requires continuous investment and deregulation so as to ensure fewer barriers to entry. Further, implementation of rural development and productivity initiatives can improve subsistence farming and agricultural practices for those who depend on these for survival.

Promoting Equality of Opportunity

Enhancing Social mobility through extending opportunities to those who were previously excluded can aid in improving current living conditions and wealth scores of citizens who live in poverty. Avenues of accomplishing such objectives can include improving early childhood development services and regional development. Education can dictate an individual's chances of acquiring employment, job promotion and higher earnings. Those who are barred from the right to education cannot benefit from income security and improved living standards.

Government investments in entrepreneurship, job creation and education, can enable the extension of employment opportunities to those who want to work, both in rural and urban regions. Further, by improving entrepreneurship opportunities, smaller firms with greater potential can also have the same opportunities to compete and integrate into global and regional value chains as larger firms. Deregulation is highly required to level the playing field in concentrated business segments.

Conclusion

The effects of the pandemic were not identical for individuals and households within different wealth populations. The divide between rich and poor translated into significant spatial differences in vulnerabilities across countries and regions. Those without access to basic amenities proved poorly lockdown prepared and therefore more vulnerable to infection. The absence of these essential services was a major component in increased vulnerability and secondary infection attack rates in low-income regions. Wealthier populations have access to clean water, safe sanitation, reliable electricity and regular income, which substantially reduces their overall vulnerability, consequently reducing their chances of becoming infected. This is untrue for low-income nations where economic, financial and health infrastructures are less-established, with limited resources available to cushion citizens from financial shocks and disasters. The wealth divide could very well mean the poor either survive or succumb to the Pandemic or future crises. The Pandemic succeeded in exposing and reinforcing socio-economic and spatial inequalities, in that not only did the expanse between rich and poor widen, but poverty, hunger and food insecurity rates have all increased as a consequence.

Spatial differences are seen in higher lockdown preparedness scores occurring in affluent, urban regions compared to rural regions, while overall COVID vulnerability was the highest in poorer, rural and underdeveloped regions and lower in high-income regions.

Across the three countries being analyzed, South Africa, Kenya and Ghana show that lockdown preparedness was inadequate among households with lower wealth status. Poor citizens could not protect themselves efficiently due to their poor living conditions. Kenya and Ghana indicated the same results showing low lockdown preparedness in rural and poor regions. Therefore, vulnerability was found to be disproportionately higher in poorer, rural and male-headed households.

Intensity of vulnerability was found to be higher in regions where average vulnerability scores were highest. Intensity was particularly higher in poorer and rural regions, where there were more household members and elderly people present.

Empirical evidence shows that income and wealth are great determinants of lockdown readiness and overall COVID-19 vulnerability. Low wealth scores and poor living conditions are associated with poorer preparedness and higher vulnerability, with regions with little access to safe water and sanitation and reliable electricity being the worst affected regions. This was the case for South Africa, Ghana and Kenya.

Policy responses need to address lack of basic service delivery, price regulations for staple foods and food subsidies, further investment in education, job creation and healthcare and improved social protection policies. These measures need to improved upon, specifically in poor and rural regions, including informal settlements in urban regions. Improving the provisions of basic needs and services in poorer regions will assist in their ability to comply with prevention and precautionary measures as stated by health bodies, thereby improving their chances of survival during health crises.

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